

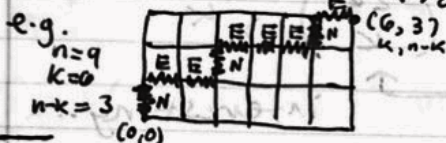
9/18

Quick review of binomial (and multinomial) coefficients

The Binomial coefficient $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ has several (easy) interpretations

e.g. $\binom{4}{2} = 6 \rightarrow$ $= \# \text{ words with } k \text{ 1's and } (n-k) \text{ 2's,}$
 $= \# \{ \underbrace{1122, 1212, 1221, 2112, 2121, 2211} \}$ i.e., rearrangements of $\underbrace{11 \dots 1}_k \underbrace{22 \dots 2}_{n-k}$

$= \# \text{ lattice paths in } \mathbb{Z}^2 \text{ taking east or north steps, from } (0,0) \text{ to } (k, n-k)$



$\leftrightarrow$ $1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9$
 $N \ E \ E \ N \ E \ E \ E \ E \ E \leftarrow \binom{9}{6}$
 (could be 1's/2's instead of E's/N's)

$= \# \text{ subsets of } [n] := \{1, 2, \dots, n\} \text{ of size } k$

e.g. $1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9$
 $N \ E \ E \ N \ E \ E \ E \ N \ E \leftrightarrow \{2, 3, 5, 6, 7, 9\} \subseteq [9]$
 (position of E's in word)

Of course, we have

Thm (Binomial Theorem) $(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$

Lemma (Pascal's Identity)

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$$

Pf (bijection):

$A \subseteq [n] \rightarrow \begin{cases} A & \text{if } n \notin A \\ A - \{n\} & \text{if } n \in A \end{cases}$

Pascal's triangle:

$n=0$	1
$n=1$	1 1
$n=2$	1 2 1
	1 3 3 1
	1 4 6 4 1
	1 5 10 10 5 1

Multinomials

How many rearrangements (anagrams) of BANANAS?

i.e., of 3A's, 1B, 2N's, 1S? (equiv. of $1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7$
 $A \ A \ A \ B \ N \ N \ S$)

∃ a transitive action of the symmetric group G_7^V of permutations of $[7]$ on the rearrangements

e.g., perm. $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 1 & 4 & 3 & 6 & 7 & 5 \end{pmatrix}$ sends $\begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ A & A & A & B & N & N & S \end{matrix} \rightarrow \begin{matrix} 2 & 1 & 4 & 3 & 6 & 7 & 5 \\ A & A & B & A & N & S & N \end{matrix}$

The stabilizer of this action is $G_3 \times G_1 \times G_2 \times G_1 \subseteq G_7$
 $G_{\{1,2,3\}} \times G_{\{4\}} \times G_{\{5,6\}} \times G_{\{7\}}$

So by Orbit-Stabilizer, # rearrangements = $\frac{\text{size of } G_7}{\text{size of stabilizer}} = \frac{7!}{3!1!2!1!} = \binom{7}{3,1,2,1}$

The Multinomial Coefficient $\binom{n}{k_1, k_2, \dots, k_m} = \frac{n!}{k_1! k_2! \dots k_m!}$ for $n = k_1 + k_2 + \dots + k_m$

= # words k_1 1's, k_2 2's, ..., k_m m's, i.e., rearrang's of $\underbrace{1 \dots 1}_{k_1} \underbrace{2 \dots 2}_{k_2} \dots \underbrace{m \dots m}_{k_m}$

= # lattice paths in $\mathbb{Z}^m$ taking steps $\vec{e}_1, \vec{e}_2, \dots, \vec{e}_m$ from $0 = (0, \dots, 0)$ to $(k_1, k_2, \dots, k_m)$
 (same correspondence between words + walks as w/ binomials)

= # chains $\emptyset = S_0 \subset S_{k_1} \subset S_{k_1+k_2} \subset \dots \subset S_{n=k_1+\dots+k_m} = [n]$
 of subsets of $[n] = \{1, 2, \dots, n\}$ for which
 $\# S_i = i \quad \forall i = 0, k_1, k_1+k_2, \dots, n$

e.g. for $\binom{7}{3,1,2,1}$ have $2131314 \leftrightarrow \emptyset \subset \{2,4,6\} \subset \{1,2,4,6\} \subset \{1,2,3,4,5,6\} \subset [7]$

Note $\binom{n}{k} = \binom{n}{n-k}$ in multinomial notation

Also note: $\binom{n}{k_1, k_2, \dots, k_m} = \binom{n}{k_1} \cdot \binom{n-k_1}{k_2} \cdot \binom{n-k_1-k_2}{k_3} \cdot \dots \cdot \binom{n-k_1-\dots-k_{m-2}}{k_{m-1}} \binom{k_m}{k_m}$

And... Multinomial Theorem: $(x_1 + x_2 + \dots + x_m)^n = \sum_{\substack{(k_1, k_2, \dots, k_m) \\ k_1+k_2+\dots+k_m=n}} \binom{n}{k_1, k_2, \dots, k_m} x_1^{k_1} x_2^{k_2} \dots x_m^{k_m}$

9/20

Permutations and cycles (Stanley § 1.3)

Recall $S_n (= \mathcal{G}_n)$ = Symmetric group on n letters
 = permutations of $[n] := \{1, 2, \dots, n\}$
 = $\{\text{bijections } \sigma : [n] \rightarrow [n]\}$

Notations • two-line $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 \\ 7 & 1 & 3 & 12 & 2 & 10 & 5 & 4 & 11 & 6 & 9 & 8 \end{pmatrix}$

• one-line $\sigma = (7, 1, 3, 12, 2, 10, 5, 4, 11, 6, 9, 8)$

digraph = directed graph

• functional digraph: $\sigma = \begin{matrix} 1 & \rightarrow & 7 \\ \uparrow & & \downarrow \\ 2 & \leftarrow & 5 \end{matrix}$ $\begin{matrix} 3 & \rightarrow & 12 \\ \uparrow & & \downarrow \\ 8 & \leftarrow & 10 \end{matrix}$ $\begin{matrix} 4 & \rightarrow & 11 \\ \uparrow & & \downarrow \\ 6 & \leftarrow & 9 \end{matrix}$

• cycle notation: $\sigma = (1\ 7\ 5\ 2)\ (3)\ (4\ 12\ 8)\ (6\ 10)\ (9\ 11)$
 $= (8\ 4\ 12)\ (10\ 6)\ (5\ 2\ 7)\ (3)\ (11\ 9)$
 $= \text{etc.} = (3)\ (7\ 5\ 2\ 1)\ (10\ 6)\ (11\ 9)\ (12\ 8\ 4)$

cycle type of σ
 = partition of sizes of cycles of σ

standard form: • each cycle has its biggest element first
 • cycles appear w/ biggest elements increasing left-to-right

Q: How many $\sigma \in S_n$ of cycle type $\lambda = (\lambda_1, \lambda_2, \dots) \vdash n$
 $= 1^{c_1} 2^{c_2} 3^{c_3} \dots$

eg. $n=4$

$\lambda = 1^4 = \begin{matrix} \square & \square & \square & \square \end{matrix}$ (a)(b)(c)(d) $\frac{4!}{1!} = 24$
 $2^1 1^2 = \begin{matrix} \square & \square & \square \end{matrix}$ (ab)(c)(d) $\frac{4!}{2!} = 6$
 $2^2 = \begin{matrix} \square & \square \end{matrix}$ (ab)(cd) $\frac{4!}{2! \cdot 2!} = 3$
 $3^1 1^1 = \begin{matrix} \square & \square & \square \end{matrix}$ (abc)(d) $2! \cdot \frac{4!}{3!} = 8$
 $4^1 = \begin{matrix} \square \end{matrix}$ (abcd) $3! \cdot \frac{4!}{4!} = 6$

multiplicity notation:

eg. $\lambda = (5, 5, 5, 3, 2, 2, 2, 1, 1)$
 $= 1^2 2^4 3^1 4^0 5^3$
 i.e. $c_1=2, c_2=4, c_3=1, c_4=0, c_5=3$

Prop: There are $\frac{n!}{1^{c_1} c_1! 2^{c_2} c_2! 3^{c_3} c_3! \dots}$ perm's in S_n
 of cycle type $\lambda = 1^{c_1} 2^{c_2} 3^{c_3} \dots$

Pf of prop: Note that S_n acts on the set of perm's with cycle type $= \lambda$ transitively, by conjugation:

e.g.,
$$\underbrace{\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ a & b & c & d & e & f & g \end{pmatrix}}_{\pi} \cdot \underbrace{[(1234)(567)]}_{\sigma \text{ of type } 4'3'} \cdot \underbrace{\begin{pmatrix} a & b & c & d & e & f & g \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{pmatrix}}_{\pi^{-1}} = (abcd)(efg)$$

9/12 So the # of such perm's = size of the orbit

$$\xrightarrow{\text{orbit stabilizer}} = \frac{|S_n|}{|Z_{S_n}(\sigma_\lambda)|} \text{ if } \sigma_\lambda \text{ is a perm. of type } \lambda$$

where $Z_{S_n}(\sigma_\lambda) := \{ \pi \in S_n : \pi \sigma_\lambda = \sigma_\lambda \pi \}$ is the centralizer of σ_λ in S_n .
 i.e. $\pi \sigma_\lambda \pi^{-1} = \sigma_\lambda$

Which perm's centralize $\sigma_\lambda = \underbrace{(a)(b) \dots}_{c_1 \text{ 1-cycles}} \underbrace{(cd)(ef) \dots}_{c_2 \text{ 2-cycles etc.}} \dots$?

- Products of powers of each cycle: there are $1^{c_1} 2^{c_2} 3^{c_3} \dots$ of these
- perm's that swap cycles of same size: there are $c_1! c_2! c_3! \dots$ of these (preserving cyclic order and biggest element)
- Products of those: $1^{c_1} c_1! 2^{c_2} c_2! \dots$ many

e.g. $\sigma_\lambda = (1234)(567)(8910)$

is centralized by $\pi = \underbrace{(4321)}_{(1234)^3} \underbrace{(59)(610)(78)}_{\text{swaps } (567) + (8910)}$

Thus $|\text{orbit}| = \frac{n!}{\prod_{j=1}^n j^{c_j} c_j!}$ as claimed. $\square$

Note: Stanley presents slightly different (but equivalent) proof by considering standard forms of perm's σ_λ .

There is an elegant reformulation of above in terms of g.f.'s:

Cor(Touchard) For $\sigma \in S_n$, let $c_k(\sigma) := \#$ of size k cycles of σ .

$$\sum_{n=0}^{\infty} \frac{1}{n!} \left(\sum_{\sigma \in S_n} t_1^{c_1(\sigma)} t_2^{c_2(\sigma)} t_3^{c_3(\sigma)} \dots \right) x^n = e^{t_1 \frac{x}{1} + t_2 \frac{x^2}{2} + t_3 \frac{x^3}{3} + \dots}$$

$$= e^{\sum_{j=1}^{\infty} t_j \frac{x^j}{j}}$$

q/24 $(\mathbb{C}[t_1, t_2, t_3, \dots])[x]$

PS: (We'll see a more conceptual proof later...)

$$e^{t_1 \frac{x}{1} + t_2 \frac{x^2}{2} + \dots} = e^{t_1 \frac{x}{1}} e^{t_2 \frac{x^2}{2}} \dots$$

$$= \left(\sum_{c_1=0}^{\infty} \frac{(t_1 x)^{c_1}}{c_1!} \right) \left(\sum_{c_2=0}^{\infty} \frac{(t_2 x^2)^{c_2}}{c_2!} \right) \dots$$

$$= \sum_{(c_1, c_2, \dots)} x^{1 \cdot c_1 + 2 \cdot c_2 + \dots} \frac{t_1^{c_1} t_2^{c_2} \dots}{1^{c_1} c_1! 2^{c_2} c_2! \dots}$$

$$= \sum_{n=0}^{\infty} x^n \frac{1}{n!} \sum_{\substack{(c_1, c_2, \dots) \\ \sum j \cdot c_j = n}} \left(\frac{n!}{1^{c_1} c_1! 2^{c_2} c_2! \dots} \right) t_1^{c_1} t_2^{c_2} \dots$$

$\hookrightarrow = \# \{ \sigma \in S_n : \begin{matrix} \text{ } \\ \sigma \text{ has } c_j \text{ } j\text{-cycles} \end{matrix} \}$
by previous prop.

$$= \sum_{n=0}^{\infty} x^n \frac{1}{n!} \sum_{\sigma \in S_n} t_1^{c_1(\sigma)} t_2^{c_2(\sigma)} \dots$$

To a chord's theorem has many important consequences, a few of which we now review:

① DEF'N $c(n, k) := \# \{ \sigma \in S_n : \sigma \text{ has } k \text{ total cycles} \}$

(Signless) Stirling
numbers of 1st kind

I.e., $\sum_{k=1}^n c(n,k) t^k = \sum_{\sigma \in S_n} t^{\# \text{cycles}(\sigma)}$

Cor (to Touchard) $\sum_{k=1}^n c(n,k) t^k = t(t+1)(t+2)\dots(t+(n-1))$

9/27 PS: Set $t_1 = t_2 = \dots = t$ in Touchard's thm to get

$$\begin{aligned} \sum_{n \geq 0} \frac{x^n}{n!} \sum_{\sigma \in S_n} t^{\# \text{cycles}(\sigma)} &= e^t \left(\frac{x^1}{1} + \frac{x^2}{2} + \frac{x^3}{3} + \dots \right) \\ &= e^{t(-\log(1-x))} \\ &= e^{\log((1-x)^{-t})} \\ &= (1-x)^{-t} \\ &= \sum_{n \geq 0} \binom{-t}{n} (-x)^n \\ &= \sum_{n \geq 0} \binom{t+n-1}{n} x^n \\ &= \sum_{n \geq 0} \frac{t(t+1)\dots(t+(n-1))}{n!} x^n \end{aligned}$$

Pf follows by comparing coeffs of $x^n/n!$. $\square$

Remark: Prop The map $S_n \rightarrow S_n$
 $\sigma \mapsto \hat{\sigma}$ = put σ in standard cycle form and erase parentheses to view in 1-line not.
 is a bijection, w/ $\# \text{cycles}(\sigma) = \# \text{left-to-right maxima in } \hat{\sigma}$.

Hence, $\sum_{\sigma \in S_n} t^{\# \text{L-to-R max.}(\sigma)} = t(t+1)\dots(t+(n-1))$

eg: $n=3$

σ	$\# \text{L-to-R max.}$
1 2 3	3
1 3 2	2
2 1 3	2
2 3 1	2
3 1 2	1
3 2 1	1

Pf: (by example)

$\sigma \mapsto \hat{\sigma}$
 $(3)(2)(1) \mapsto 3:2:1$
 $(3)(2)(1)(4)(6) \mapsto 3:2:1:4:6$
 is reversible. Just put (before each L-to-R maxima, and put) right before the end and at the end. $\square$

(2) (Cor of Touchard) Can compute $E_k(n) :=$ expected # of k-cycles in a uniformly random $\sigma \in S_n$

$$E_k(n) = \frac{1}{n!} \sum_{\sigma \in S_n} c_k(\sigma) = \frac{1}{n!} \left[\frac{2}{k} \sum_{\sigma \in S_n} t_1^{c_1(\sigma)} t_2^{c_2(\sigma)} \dots \right]_{t_1=t_2=\dots=1}$$

$$\text{So } \sum_{n \geq 0} E_k(n) x^n = \left[\frac{2}{k} \sum_{n \geq 0} \frac{x^n}{n!} \sum_{\sigma \in S_n} t_1^{c_1(\sigma)} t_2^{c_2(\sigma)} \dots \right]_{t_1=t_2=\dots=1}$$

$$= \left[\frac{2}{k} e^{t_1 \frac{x^1}{1} + t_2 \frac{x^2}{2} + t_3 \frac{x^3}{3} + \dots} \right]_{t_1=t_2=\dots=1}$$

$$= \left[\frac{x^k}{k} e^{t_1 \frac{x^1}{1} + t_2 \frac{x^2}{2} + \dots} \right]_{t_i=1}$$

$$= \frac{x^k}{k} e^{x^1 + \frac{x^2}{2} + \dots} = \frac{x^k}{k} e^{-\log(1-x)} = \frac{x^k}{(1-x)}$$

$$= \sum_{n \geq k} \frac{1}{k} x^n \Rightarrow E_k(n) = \begin{cases} 1/k & \text{if } n \geq k, \\ 0 & \text{otherwise.} \end{cases}$$

Note: $E_k(n)$ eventually constant in n . In fact, one can show #k-cycles of random $\sigma \in S_n$ converges (as $n \rightarrow \infty$) to a Poisson random variable w/ expectation $\lambda = 1/k$.

(3) (Cor to Touchard) There are special classes of perms defined by restrictions on their cycle sizes, so all have nice g.f.'s.

e.g.: no large cycles

$\sigma \in S_n$ is an involution (i.e. $\sigma^2 = e$)

$\Leftrightarrow \sigma$ has only 1- and 2-cycles
($\sigma = (ab)(cd)\dots$)

$$\text{So } \sum_{n \geq 0} \frac{x^n}{n!} \# \{ \text{involutions } \sigma \in S_n \} = \left[e^{t_1 \frac{x^1}{1} + t_2 \frac{x^2}{2} + \dots} \right]_{t_1=t_2=\dots=1}$$

$$= e^{x + \frac{x^2}{2}}$$

and even $\sum_{n \geq 0} \frac{x^n}{n!} \sum_{\substack{\sigma \in S_n \\ \text{involution}}} t_1^{\# 1\text{-cycles}(\sigma)} = e^{tx + x^2}, \text{ etc.}$

What about no small cycles?

DEF 'N A derangement $\sigma \in S_n$ is a perm. w/ no fixed points,
i.e., w/ $\# \{i \in [n] : \sigma(i) = i\} = 0$.

Q: (Derangement / Hat-check problem): $n \geq 100$ people check their hats; attendant gives people's hats back randomly; what is prob. that no one gets their ^{own} hat back?

9/29

i.e., what is $\frac{d_n}{n!}$, where $d_n := \# \{ \sigma \in S_n : \sigma \text{ is a derangement} \}$?

$$\begin{aligned} \sum_{n \geq 0} \frac{x^n}{n!} d_n &= \left[e^{t_1 x + t_2 x^2 + \dots} \right]_{t_1=0, t_2=0, t_3=\dots=1} \\ &= e^{x^2/2 + x^3/3 + \dots} \\ &= e^{-\log(1-x) - \frac{x}{1}} = \boxed{\frac{e^{-x}}{1-x}} \end{aligned}$$

$$\begin{aligned} \text{But } \frac{e^{-x}}{1-x} &= (1 + x + x^2 + \dots) \left(1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots \right) \\ &= \sum_{n \geq 0} x^n \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + (-1)^n \frac{1}{n!} \right) \end{aligned}$$

$$\begin{aligned} \text{So } \frac{d_n}{n!} &= 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + (-1)^n \frac{1}{n!} \\ &\xrightarrow[\text{converges rapidly as } n \rightarrow \infty]{} e^{-1} = 1/e \approx 0.368 \dots \end{aligned}$$

Note: (in fact, $d_n =$ closest integer to $\frac{n!}{e}$ $\forall n$.)

Also have recurrence for d_n :

$$\text{Prop. } d_n = (n-1) \cdot (d_{n-1} + d_{n-2})$$

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